HKU MAIB7002 | GROUP Q | FRESH RETAIL

FRESH FOOD. SMARTER DISCOUNTS.

Should we discount it tomorrow?

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7.9% **lower prediction error**

RF + CatBoost (50/50) vs. 7-day average

Final-period MAE: 0.2892 vs. 0.3142 | 2,184 store-product-days

① Forecast tomorrow's sales

Help a Dingdong category manager compare daily discounts. Public FreshRetailNet histories contain sales, price rates, stockouts and calendar inputs.

50,000 public histories → **16,684** eligible histories → **312** in five stores

Keep 15-85% discounted days (rate < 0.95), 6+ discount-state changes and <20% zero-sales days; choose the five stores with most eligible histories.

312 histories / 122 products / one city

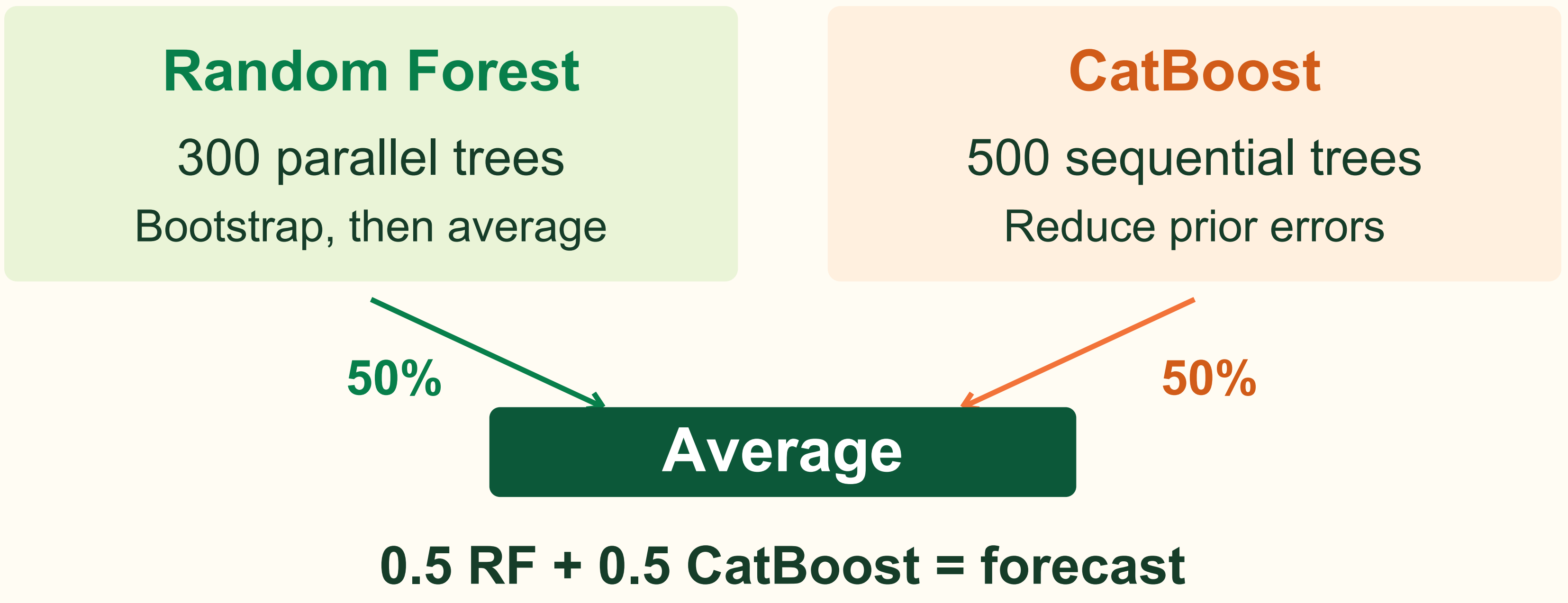
Fit: 4 Apr - 25 Jun 2024; 25,896 rows

Validation: 5 expanding weekly folds

Benchmark: 26 Jun - 2 Jul; rolling one day ahead

② Combine two learners

10 history, calendar and promotion inputs.
RF: squared-error splits; CatBoost: RMSE loss.
Target: normalized observed sales, not units.



Random Forest
300 parallel trees
Bootstrap, then average

CatBoost
500 sequential trees
Reduce prior errors

50% 50%

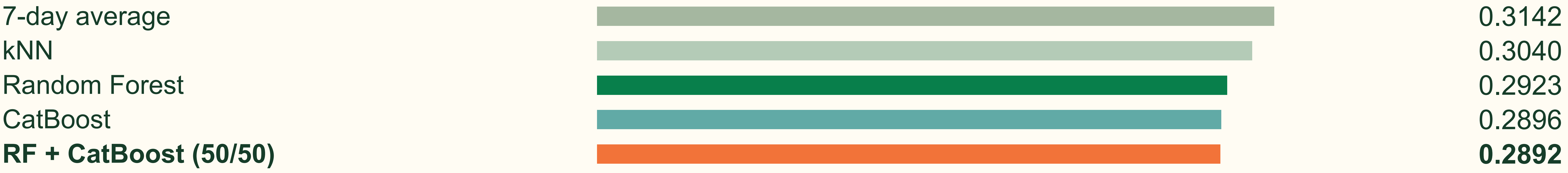
Average

0.5 RF + 0.5 CatBoost = forecast

RF: depth 10, leaf 3 | CB: depth 5, learning rate 0.03

③ How much does prediction improve?

MAE = mean absolute error; lower is better. Same final-period cases.



Ensemble beats the baseline in all 5 validation weeks. Final RMSE (root mean square error): 0.4593.

④ Why a 50/50 blend?

RF / CatBoost (%)	Validation MAE
0 / 100	0.30330
40 / 60	0.30144
50 / 50	**0.30125**
60 / 40	0.30118
100 / 0	0.30244

60/40 is marginally lower. We retain 50/50 for simplicity on a nearly flat validation curve.

⑤ Where errors remain

Observed rate	MAE	Cases
Full/light (>0.95)	0.2330	1,186
Moderate (0.80, 0.95]	0.2982	757
Deep (<=0.80)	0.5378	241

Deep-discount MAE is 2.3x full/light-rate MAE.

⑥ From a forecast to a discount

5% off

Store 18 / Product 11 / 26 Jun 2024

Predicted sales: 1.12 at full price -> 1.26

Value proxy: 0.95 x 1.26 = 1.20, vs. 1.12 at full price

Illustrative scenario; no observed uplift. Zero unit cost assumed.

Compare eight price rates; require 3+ training days within 0.025 of each candidate rate. Apply assumed stock rules, then maximize **price rate x forecast sales.**

Better forecasts. Pricing impact still to test.

Discounts are observational; sales are stock-limited. No measured profit or waste benefit.
Retrospective cohort and previously viewed benchmark. Next: inventory/cost data and a controlled pilot.

FreshRetailNet-50K / Dingdong / CC BY 4.0. Normalized sales. Final notebook: pricing-ml-ap @ a7e26dd.

CODE + NOTEBOOK